Safe Reinforcement Learning via Minmax Penalties

School of Computer Science & Applied Mathematics
University of the Witwatersrand

Wendy Maboa
2541693

Supervised by Geraud Nangue Tasse

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Abstract

Reinforcement Learning (RL) has demonstrated remarkable results, from mastering games like Atari and Go to shaping the behaviour of Large Language Models (LLMs). Existing safety alignment frameworks, including KL divergence penalties and learned cost models, address this challenge but introduce annotation costs, additional hyperparameters, or resource-intensive components. Moreover, models fine-tuned with these approaches can still produce harmful content such as hate speech and discriminatory outputs. This research proposes that a safety signal may already exist within the RL loop itself. Drawing from the ROSARL algorithm, we adapt its Minmax penalty to the LLM fine-tuning setting, replacing both the KL divergence penalty and the learned cost model with a value-function-derived safety signal. This eliminates the need for a separate cost model or Lagrange multiplier. The proposed experiments evaluate the approach on GPT-2 small and a small Qwen model (exact size in the 0.5 to 3B parameter range, subject to available compute), trained on BeaverTails with Detoxify as the unsafe-state detector, against a KL-penalty baseline on BeaverTails-Evaluation. This research provides an accessible route to improving safety in PPO-based LLM fine-tuning without additional annotation or hyperparameter tuning.

Declaration

I, Wendy Maboa, hereby declare the contents of this proposal to be my own work. This proposal is submitted for the degree of Master of Science in the field of Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree.

Wendy Maboa
July 19, 2026

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4. Data Analysis (presentation, coding, interpretation) [ ]
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Student number: 2541693

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Contents

Chapter 1

Introduction

RL has a history of exceeding its own expectations. For decades, its successes were confined to toy domains and simple simulations. That changed in 2015 when Mnih *et al.* [2015] showed that a single neural network, trained with RL from raw pixel inputs, could reach human-level performance across dozens of Atari games without any hand-engineered features. A year later, AlphaGo [Silver *et al.* 2016] defeated the European Go champion, solving a combinatorial problem thought to be decades from machine mastery. These milestones did more than set benchmarks: they established that deep RL could discover sophisticated strategies in high-dimensional spaces, and shifted the field's question from *whether* RL could match human performance to *which domains* it would transform next.

The answer, increasingly, is language. Christiano *et al.* [2017] showed that complex RL tasks could be solved from pairwise human preference labels without access to a true reward function. Ouyang *et al.* [2022] scaled this insight to GPT-3, producing InstructGPT: a model that substantially outperformed its base counterpart on instruction-following by treating the LLM as an RL agent. In the RLHF paradigm, an incoming prompt is the state, token-by-token generation is a sequence of actions, and a reward model trained on human preferences provides the scalar feedback signal. This framing has become the dominant paradigm for aligning LLMs with human intent [Lambert *et al.* 2023].

Interestingly, this RL framing of language model alignment is not just a computational convenience. Neuroscience research on harm avoidance in biological agents offers a complementary perspective: Lockwood *et al.* [2020] showed that when learning to avoid harming others, humans prioritise model-free RL signals, the computationally efficient system that assigns values to actions through accumulated experience rather than deliberate reasoning. The *NeuroAI for AI Safety* roadmap [Orban and others 2024] formalises this intuition, arguing that studying how biological agents navigate environments while maintaining self-preservation is a productive route toward AI safety. The Minmax penalty proposed in this work operates on precisely this principle: a simple, reactive, value-based signal that does not require explicit reasoning about harm categories.

Yet RLHF has also introduced a new class of safety challenge. LLMs fine-tuned with RLHF can and do produce harmful content, including hate speech, discriminatory outputs, and instructions for self-harm, at rates that undermine their trustworthiness in deployment. The established toolkit for constrained optimisation in RL [García and Fernández 2015; Achiam *et al.* 2017] has been adapted to address this, but the resulting approaches carry significant costs. KL divergence penalties constrain how far the policy drifts from a reference model, requiring manual tuning and introducing an implicit tradeoff between safety and capability. Learned cost models [Dai *et al.* 2024] train a separate model on human harmlessness labels and balance it against the helpfulness reward via a Lagrange multiplier. HC-RLHF [Chittepu and others 2025] adds high-confidence probabilistic guarantees but inherits the same infrastructure requirements. All of these approaches add components external to the core RL loop, introducing annotation cost, hyperparameter burden, or both. Figure 1.1 summarises the structural differences.

Approach	**Extra model?**	**Extra annotation?**	**New hyperparam?**	**Online?**
KL baseline (standard RLHF)	No	No	Yes (β)	Yes
Safe RLHF / CPO	Yes	Yes	Yes (λ, d)	Yes
HC-RLHF	Yes	Yes	Yes	Yes
Constrained DPO	No	No	No	No
PPO + Minmax (ours)	**No**	**No**	**No**[1]	**Yes**

Figure 1.1: Structural comparison of safe RLHF approaches. The Minmax penalty is the only online method requiring no extra model, no extra annotation, and no new safety-specific hyperparameters.

This raises the central question of this proposal: can we replace KL divergence penalties and learned cost models with a theoretically motivated safety signal derived directly from the agent's own value function, without introducing new hyperparameters or auxiliary models?

The ROSARL algorithm [Nangue Tasse and others 2026] answers this in the affirmative for robotic control tasks. At each step the agent maintains running estimates of the smallest and largest rewards it has observed, $R_{\min}$ and $R_{\max}$, and running value bounds $V_{\min}$ and $V_{\max}$ updated jointly from the reward bounds and the critic's current value estimate $V(s_t)$:

$$R_{\min} \leftarrow \min(R_{\min}, r_t), \quad R_{\max} \leftarrow \max(R_{\max}, r_t), \tag{1.1}$$

$$V_{\min} \leftarrow \min(V_{\min}, R_{\min}, V(s_t)), \quad V_{\max} \leftarrow \max(V_{\max}, R_{\max}, V(s_t)). \tag{1.2}$$

[1]The Minmax penalty itself introduces no new hyperparameters: $V_{\min}$ and $V_{\max}$ are derived from the agent's own training trajectory. The KL anchor ($\beta = 0.01$) retained in the implementation is a conservatism term inherited from the PPO training loop to prevent mode collapse, not a safety-specific parameter. The unsafe detection threshold τ is a property of the detector (Detoxify), not of the Minmax mechanism.

When the unsafe-state detector fires, the reward at that step is replaced with

$$R_{\text{unsafe}} = V_{\min} - V_{\max}, \tag{1.3}$$

which is guaranteed non-positive and is the worst possible value any reachable state can take under the observed reward range and value estimates. $V_{\min}$ and $V_{\max}$ are dynamic running estimates that update every step via the three-way rule in (1.2); see Algorithm 1 in Chapter 4 for the full update rule. As more of the reward landscape and value range is explored, R_{unsafe} becomes monotonically more negative, providing increasingly strong corrective signal without any manual tuning.

This proposal also connects to the emerging paradigm of RL with verifiable rewards [Lambert and others 2024; Youssef and others 2025; Su and others 2025], where reward signals are computed deterministically rather than approximated by a learned model. The Minmax penalty fits this paradigm: R_{unsafe} is derived directly from critic estimates, not from an opaque learned cost function. Where recent work asks whether verifiable rewards preserve safety alignment [Huang and others 2025], this proposal investigates the complementary direction: using a value-function-derived signal to actively *improve* it.

Because Algorithm 1 only acts when the unsafe-state detector fires, detector quality is expected to be the primary variable affecting coverage. This motivates a two-phase research programme. In Phase 1, Detoxify [Hanu and Unitary team 2020] serves as the unsafe-state detector, providing a fast, reproducible safety signal for harm categories that manifest as surface-level toxic language. In Phase 2, Detoxify is replaced by an activation-based probe [Rateike *et al.* 2023; Han *et al.* 2025], enabling Algorithm 1 to intercept safety-relevant states before the output token is produced, and covering categories that output-level classifiers miss, such as self-harm. This two-phase architecture is illustrated in Chapter 4.

The contributions of this study are as follows.

1. **Contribution 1.** First application of the ROSARL Minmax penalty to PPO-based LLM fine-tuning. The study will evaluate this under multiple base models (GPT-2 small and a Qwen model at 1 to 3B parameters) and fine-tuning strategies (full fine-tuning and LoRA [Hu *et al.* 2022]), isolating the benefit to the Minmax mechanism rather than any specific model or training configuration.

2. **Contribution 2.** A proposed bridge between Algorithm 1 and activation-based safety detection [Han *et al.* 2025; Rateike *et al.* 2023]. Following recent follow-up work to Rateike *et al.* [2023] (currently under review) identifying hallucination detection as the most tractable initial subset for activation-level probes, the probe will first target factually incorrect and internally inconsistent outputs before extending to broader harm categories.

3. **Contribution 3.** A theoretical and empirical analysis of the Minmax penalty as a verifiable safety signal, situating it within the RLVR literature [Lambert and others 2024; Youssef and others 2025] and the safety-capability tradeoff debate [Huang and others 2025].

Chapter 2 provides background and related work. Chapter 3 states the research questions and hypotheses. Chapter 4 describes the methodology. Chapter 5 outlines the research plan. Chapter 6 concludes.

Chapter 2

Background and Related Work

This chapter reviews the technical foundations required to situate the proposed research. Section 2.1 covers RL fundamentals and the RLHF pipeline. Section 2.2 reviews the safe RL literature. Section 2.3 introduces the RLVR paradigm. Section 2.4 describes the ROSARL algorithm in detail. Section 2.5 covers the datasets used. Section 2.6 reviews activation-based safety detection, which forms the basis of Phase 2.

2.1 Reinforcement Learning and RLHF

RL models sequential decision-making as a Markov Decision Process (MDP), defined as a tuple (S, A, P, R, γ), where S is the state space, A the action space, P the transition dynamics, R the reward function, and γ the discount factor. The agent learns a policy π maximising expected cumulative discounted reward [Puterman 2014; Sutton and Barto 2018]:

$$J(\pi) = \mathbb{E}_{\tau \sim \pi}\left[\sum_{t=0}^{T} \gamma^t R(s_t, a_t)\right] \tag{2.1}$$

PPO [Schulman *et al.* 2017] is the standard RL algorithm for RLHF. It stabilises training via a clipped surrogate objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(r_t(\theta)\,\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\Big)\Big] \tag{2.2}$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the probability ratio and $\hat{A}_t$ is the advantage estimate. PPO maintains separate actor and critic networks; the critic provides value estimates $V(s)$ that are central to Algorithm 1.

The modern RLHF pipeline trains a reward model on human preference labels and uses PPO to fine-tune the LLM to maximise this reward subject to a KL divergence penalty:

$$R_{\text{RLHF}}(s, a) = r_\phi(s, a) - \beta\,\text{KL}[\pi_\theta(\cdot|s)\,\|\,\pi_{\text{ref}}(\cdot|s)] \tag{2.3}$$

where r_ϕ is the reward model, π_{ref} is the frozen reference policy, and β is the KL coefficient. This KL term, first introduced by Ziegler *et al.* [2019] and scaled to production by Ouyang *et al.* [2022] with InstructGPT, is the primary safety mechanism in standard RLHF: a regulariser rather than a principled safety guarantee. The limitations and historical risks of this framing are discussed by Lambert *et al.* [2023], including the tendency of reward models to encode annotation artefacts, and the lack of a principled basis for setting β.

Safe RLHF [Dai *et al.* 2024] extends this with a separately trained cost model and Lagrangian constrained optimisation. A complementary line of work, Constitutional AI [Bai *et al.* 2022], scales safety alignment by replacing some human feedback with AI-generated critiques and revisions guided by a written set of principles, reducing annotation cost at the price of a second model in the training loop:

$$\max_{\pi} J_R(\pi) \quad \text{s.t.} \quad J_C(\pi) \leq d \tag{2.4}$$

where $J_C(\pi)$ is the expected cost from the cost model and d is a safety threshold. HC-RLHF [Chittepu and others 2025] adds high-confidence constraint satisfaction, representing the current state of the art. Both require multi-GPU infrastructure, a second trained model, and annotated harmlessness labels.

2.2 Safe Reinforcement Learning

Safe RL [García and Fernández 2015] addresses the challenge of learning policies that maximise reward while satisfying behavioural constraints during training and deployment. Figure 2.1 illustrates the main families of approaches.

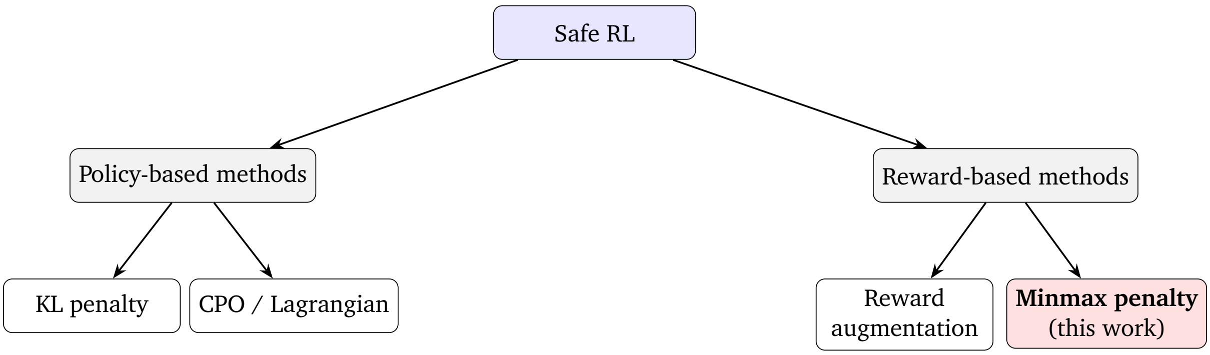

Figure 2.1: Taxonomy of safe RL approaches [García and Fernández 2015]. The Minmax penalty is a reward-based method requiring no learned cost model.

Constrained Policy Optimisation [Achiam *et al.* 2017] provides the first policy gradient method with formal worst-case constraint guarantees:

$$\pi_{k+1} = \arg\max_{\pi} J_R(\pi) \quad \text{s.t.} \;\; J_{C_i}(\pi) \leq d_i \; \forall i, \quad D_{\mathrm{KL}}(\pi \| \pi_k) \leq \delta \tag{2.5}$$

Safe RLHF adapts CPO to the RLHF setting with a learned cost model. Distributional approaches [Yang *et al.* 2022] further strengthen safety by controlling tail-risk via CVaR. These methods provide the theoretical comparators that motivate asking whether any constraint model is necessary at all.

2.3 Reinforcement Learning with Verifiable Rewards

A parallel strand of recent work reframes reward design around verifiability: using signals that can be checked exactly rather than approximated by a learned model [Lambert and others 2024; Youssef and others 2025; Su and others 2025]. A prominent example is Group Relative Policy Optimisation (GRPO), introduced by Shao *et al.* [2024] in DeepSeekMath, which replaces the learned value baseline with group-relative advantages computed from verifiable mathematical rewards. This eliminates reward model error and reduces reward hacking. Huang and others [2025] study whether such training breaks safety alignment, finding that KL constraints can preserve safety guardrails. This proposal investigates the complementary direction: using a deterministically computed, verifiable value-function signal to actively improve safety rather than merely preserve it.

2.4 The ROSARL Algorithm

ROSARL [Nangue Tasse and others 2026] proposes that the agent's own value function contains sufficient information to construct a principled safety penalty without auxiliary models. At each training step the algorithm maintains $R_{\min}$ and $R_{\max}$ as the running minimum and maximum of all rewards observed so far, and updates value bounds $V_{\min}$ and $V_{\max}$ via a three-way minimum and maximum that incorporates both the running reward bounds and the critic's current value estimate $V(s_t)$: $V_{\min} \leftarrow \min(V_{\min}, R_{\min}, V(s_t))$ and $V_{\max} \leftarrow \max(V_{\max}, R_{\max}, V(s_t))$. This anchoring ensures $V_{\min}$ tracks the worst value the agent could plausibly experience and $V_{\max}$ tracks the best, so that the penalty gap grows as the agent discovers the full range of achievable values. When the unsafe-state detector z_t fires, the reward is replaced with $R_{\text{unsafe}} = V_{\min} - V_{\max}$, which is always non-positive and lower than the value of any reachable state under the observed reward and value range. As the agent encounters wider extremes, R_{unsafe} becomes monotonically more negative, making the penalty more severe without any manual tuning. Figure 2.2 illustrates this self-calibration property.

ROSARL was validated on Safety-Gymnasium benchmarks (PointGoal, PointPush, HumanoidVelocity, PILLAR) and matches or outperforms Lagrangian, CPO, and Sauté-RL baselines under environmental stochasticity. This proposal is the first application to LLM fine-tuning.

2.5 Datasets

This section describes the datasets used for training and evaluation in this research.

BeaverTails [Ji *et al.* 2023] is a large-scale human preference dataset covering 14 harm categories: animal abuse, child abuse, controversial topics, discrimination, drug abuse, financial crime, hate speech, human rights violations, non-violent unethical behaviour, privacy violations, sexually explicit content, terrorism, violence, and weapons. Uniquely, it provides separate annotations for helpfulness and harmlessness for the

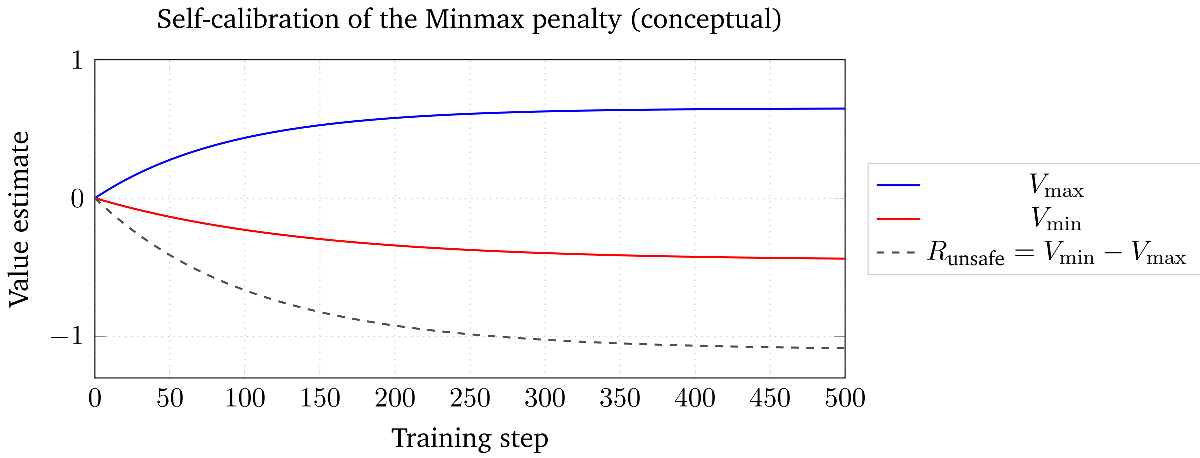

Figure 2.2: Conceptual illustration of the self-calibration property. As training progresses R_{unsafe} becomes monotonically more negative, providing increasingly strong corrective signal without any manual tuning.

same prompt. The training split (27,186 prompts) is used for fine-tuning; the held-out BeaverTails-Evaluation split (700 prompts) serves as the primary evaluation benchmark, shared with Safe RLHF and HC-RLHF to enable direct SOTA comparison.

PKU-SafeRLHF [Ji *et al.* 2024] extends BeaverTails with multi-level safety labels across 19 harm categories and three severity levels, providing paired safe and unsafe responses for the same prompts. This structure is directly relevant to the Phase 2 extension: the paired responses provide explicit positive and negative training examples for the activation probe.

HarmBench [Mazeika *et al.* 2024] provides over 500 unique adversarial harmful behaviours designed to probe LLM safety under worst-case conditions. Unlike BeaverTails, which covers naturalistic prompts, HarmBench is specifically designed for red-teaming: it includes standard, contextual, and multi-turn behaviours, as well as attacks that span multiple harm categories simultaneously. In this research, HarmBench serves as an out-of-distribution robustness evaluation, testing whether the Minmax penalty generalises beyond the harm categories and prompt styles seen during BeaverTails training. Full HarmBench evaluation is planned as a later milestone pending larger model access.

2.6 Activation-Based Safety Detection

This section reviews the activation-based safety detection literature that motivates Phase 2 of this research.

Rateike *et al.* [2023] demonstrate that LLMs encode unsafe information in internal activation vectors during generation, before the output token is produced. Using DeepScan (subset scanning), they identify specific neuron subsets in the final layers of OPT-6.7B that are anomalously active when the model is about to hallucinate. Crucially, hallucination detection was found to be the most tractable starting subset for detection, a finding from recent follow-up work currently under review that directly informs the Phase 2 design: the activation probe will target factually incorrect outputs first before extending to broader harm categories. Cintas *et al.* [2025] extend this to persona representations in decoder-only LLMs, finding that safety-relevant representations are localised in the final third of layers, with the open question of whether these activations can steer generation.

SafeSwitch [Han *et al.* 2025] provides the most direct empirical precedent for Phase 2: internal activation signals can directly steer unsafe behaviour in deployed LLMs, achieving large reductions in harmful outputs while preserving utility. SafetyNet [Chaudhary and Barez 2025] provides complementary evidence by monitoring deceptive behavioural signals in activations in real time. Nguyen *et al.* [2025] further show that distributional surgery on activations can both detect and mitigate undesirable content layerwise.

Chapter 3

Research Questions

This chapter states the problem addressed by the research and the questions and hypotheses that structure the experimental programme.

3.1 Problem Statement

LLMs fine-tuned with PPO can produce harmful outputs at deployment at rates that undermine their trustworthiness in safety-critical applications [Dai *et al.* 2024; Chittepu and others 2025]. Current mitigation strategies each introduce a structural cost: KL divergence penalties require manual tuning of the KL coefficient and trade safety against capability without a principled basis for setting the trade-off; learned cost models require a second trained model, additional harmlessness annotations, and a Lagrangian multiplier whose update dynamics introduce further hyperparameter burden. These mechanisms add components external to the core PPO loop, increasing engineering and annotation cost without providing a principled safety guarantee. This research addresses the question of whether the value function already present in the PPO training loop can itself supply a self-calibrating safety signal that replaces both mechanisms. The methodology in Section 4.1 operationalises this question through Algorithm 1, ablation studies that isolate the Minmax mechanism, and a two-phase pipeline that varies only the unsafe-state detector.

3.2 Research Questions

This research investigates whether the ROSARL Minmax penalty can serve as the primary safety mechanism in PPO-based LLM fine-tuning, replacing KL divergence penalties and learned cost models, across multiple base models and fine-tuning strategies. The experimental programme is structured around three hypotheses; each hypothesis maps directly to one dimension of the ablation design described in Section 4.1.3, so that the three together cover the safety mechanism, the detector, and the architectural extension to activation-level detection.

- **H1 (mechanism).** A policy trained with the Minmax penalty produces fewer harmful outputs than a policy trained with KL divergence alone, measured by harm rate on BeaverTails-Evaluation. This effect is expected to persist across base models (GPT-2 small and a small Qwen model) and fine-tuning strategies (full fine-tuning and LoRA), isolating the mechanism as the source of the benefit rather than any model-specific property.
- **H2 (detector quality bounds coverage).** The Minmax penalty's harm reduction is bounded by the recall of the unsafe-state detector. Categories where the detector signal is weak or absent, such as self-harm under Detoxify, are not expected to improve and may regress, because Algorithm 1 only acts when $z_t = 1$. This frames detector quality as the primary variable modulating coverage, not the penalty mechanism itself.
- **H3 (consequence of H2).** If H2 holds, then replacing Detoxify with an activation-level probe should extend coverage to categories where Detoxify's recall is near zero (most notably self-harm), without altering the underlying Minmax mechanism. H3 tests this prediction directly: the penalty is held fixed, only the source of z_t is varied, so that any change in coverage is attributable to the detector rather than to a different safety algorithm.

Chapter 4

Methodology

This chapter describes the full experimental design, implementation, evaluation strategy, ablation studies, and limitations.

4.1 Experimental Methodology

This chapter details the experimental methodology. PPO is used for all fine-tuning runs, comparing a KL-penalty baseline against the Minmax penalty. The unsafe-state detector is Detoxify in Phase 1 and an activation-based probe in Phase 2. Evaluation is performed on two benchmarks: BeaverTails-Evaluation [Ji *et al.* 2023] as the in-distribution harm-rate measure, and HarmBench [Mazeika *et al.* 2024] as an out-of-distribution adversarial robustness measure. Ablations isolate the effect of the Minmax mechanism from confounds such as base model, fine-tuning strategy, and detector threshold. The methodology is structured around four components: data collection (Section 4.1.1), the two-phase pipeline architecture (Section 4.1.2), ablation studies (Section 4.1.3), and limitations and constraints (Section 4.1.4).

4.1.1 Data Collection

To compare the Minmax penalty with existing methods, PPO fine-tuning with the Minmax penalty is implemented on GPT-2 small and a small Qwen model. The training split of BeaverTails is used for fine-tuning, with evaluation on the held-out BeaverTails-Evaluation split and on HarmBench.

Implementation framework. The primary implementation framework is the TRL library's `PPOTrainer` [von Werra *et al.* 2020], into which the Minmax penalty is integrated as a reward wrapper following the structure of Algorithm 1. The PKU-SafeRLHF codebase [Ji *et al.* 2024] is used as the reference implementation for understanding the Safe RLHF baseline design, not as the running framework: it requires DeepSpeed, which does not build on the available Windows hardware. A later milestone will port the Minmax wrapper into the PKU-SafeRLHF codebase on a Linux cluster to enable direct head-to-head comparison with the published Safe RLHF and HC-RLHF baselines under matched infrastructure.

Algorithm 1 Implementation. The Minmax penalty is implemented as a reward wrapper around the TRL `PPOTrainer` [von Werra *et al.* 2020]. The full procedure is given in Algorithm 1.

Algorithm 1 PPO + Minmax Penalty (LLM adaptation of ROSARL [Nangue Tasse and others 2026])

Require: LLM π_θ, critic V_ψ, unsafe detector $\mathcal{D}$, threshold τ, KL anchor β

Initialise $R_{\min} \leftarrow 0,\ R_{\max} \leftarrow 0,\ V_{\min} \leftarrow 0,\ V_{\max} \leftarrow 0$ ▷ as in original ROSARL

for each training step t **do**

    Sample prompt $x \sim \mathcal{X}_{\text{train}}$

    Generate response $y \sim \pi_\theta(\cdot \mid x)$

    $\delta_t \leftarrow \mathcal{D}(y)$ ▷ detector score

    $r_t^{\text{raw}} \leftarrow -\delta_t$ ▷ raw reward (LLM adaptation: negated Detoxify score)

    $V_t \leftarrow V_\psi(x, y)$ ▷ critic value estimate

    $R_{\min} \leftarrow \min(R_{\min}, r_t^{\text{raw}});\quad R_{\max} \leftarrow \max(R_{\max}, r_t^{\text{raw}})$ ▷ ROSARL reward bounds

    $V_{\min} \leftarrow \min(V_{\min}, R_{\min}, V_t);\quad V_{\max} \leftarrow \max(V_{\max}, R_{\max}, V_t)$ ▷ ROSARL three-way value bounds

    if $\delta_t > \tau$ **then** ▷ unsafe state detected

        $r_t \leftarrow \max(V_{\min} - V_{\max},\ -1.0)$ ▷ Minmax penalty; -1.0 floor is a stabilisation modification, not in original ROSARL

    else

        $r_t \leftarrow r_t^{\text{raw}}$ ▷ LLM adaptation: detector reward (original uses environment reward)

    end if

    Update π_θ, V_ψ via PPO with r_t and KL penalty β

end for

Key design choices, with explicit attribution to either ROSARL or this work:

- **Reward-bound tracking ($R_{\min}$, $R_{\max}$) and the $R_{\text{unsafe}} = V_{\min} - V_{\max}$ formula** are taken unchanged from ROSARL [Nangue Tasse and others 2026].
- **Three-way value-bound update.** Following Algorithm 1 of Nangue Tasse and others [2026], $V_{\min}$ and $V_{\max}$ are updated using a three-way minimum and maximum respectively: $V_{\min}$ incorporates both the running minimum reward $R_{\min}$ and the critic's current value estimate $V(s_t)$, ensuring the penalty gap grows as the agent discovers the full range of achievable values. $V_{\max}$ is correspondingly anchored to the highest reward and value seen, providing an optimistic upper bound that makes R_{unsafe} conservative by construction.
- **The -1.0 floor on the unsafe reward** is not part of the original Algorithm 1; it was introduced as an engineering modification to guard against catastrophic PPO ratio violations during early training, when value estimates are not yet well-calibrated and $V_{\min} - V_{\max}$ can briefly take very large negative values.

- **The safe-branch reward** $r_t \leftarrow -\delta_t$ is this work's LLM-setting adaptation: in the absence of a dense environment reward signal, the negated Detoxify toxicity score provides a continuous signal that the policy can optimise, encouraging progressively less toxic outputs even when no unsafe flag is triggered. This differs from the original Algorithm 1, in which the environment reward is passed through unchanged for safe transitions.
- **The small KL anchor (**$\beta = 0.01$**)** is retained solely to prevent mode collapse and is distinct from the $\beta = 0.2$ used in the KL baseline condition.
- **Reducing PPO epochs from 4 to 1** prevents the probability ratio from exploding when the reward sign flips abruptly between the safe and unsafe branches.

Base models and fine-tuning strategies. GPT-2 small (117M parameters) serves as the primary base model, feasible on the available hardware. A Qwen model in the 1 to 3B parameter range is planned for the model-scale ablation (Ablation 2), to be run on a compute cluster. LoRA [Hu *et al.* 2022] is used as the fine-tuning strategy for larger models, as it reduces memory requirements while still permitting comparison across fine-tuning methods in Ablation 3. In Phase 1, Detoxify [Hanu and Unitary team 2020] provides the binary signal z_t indicating whether the model output is classified as harmful. In Phase 2, Detoxify is replaced with an activation-based probe targeting hallucination detection as the initial subset, following the methodology of Rateike *et al.* [2023].

Evaluation benchmarks. The primary evaluation benchmark is the held-out BeaverTails-Evaluation set (700 prompts across 14 harm categories). Both trained models generate 32 tokens per prompt; outputs are classified as harmful or benign, and harm rate is reported in aggregate and by category, enabling direct comparison with published Safe RLHF [Dai *et al.* 2024] and HC-RLHF [Chittepu and others 2025] results, which use the same split. The secondary benchmark is HarmBench [Mazeika *et al.* 2024], which provides over 500 adversarial behaviours specifically designed for red-teaming, spanning standard, contextual, and multi-turn attacks that cross harm categories. Evaluation on HarmBench tests whether the Minmax penalty generalises beyond the naturalistic BeaverTails prompts seen during training. Following Mazeika *et al.* [2024], attack success rate (ASR) is the primary metric, reported overall and broken down by attack family. HarmBench evaluation is planned as a later phase, after Phase 1 results are stable.

4.1.2 Pipeline Implementation: Two-Phase Architecture

The research is structured in two phases, each defined by the unsafe-state detector that feeds the binary signal z_t into Algorithm 1. In both phases the PPO training loop and the Minmax penalty itself are identical; only the source of z_t changes. The two phases are illustrated in Figure 4.1 and Figure 4.2.

Phase 1: Detoxify as the unsafe-state detector. At each training step a prompt x is drawn from BeaverTails and the policy π_θ generates a response y. Detoxify [Hanu and

Unitary team 2020] scores y and returns a toxicity probability $\delta_t \in [0, 1]$; the binary signal is z_t. The Minmax wrapper then sets the per-trajectory reward to $R_{\text{unsafe}} = V_{\text{min}} - V_{\text{max}}$ whenever $z_t = 1$, and to the standard reward $-\delta_t$ otherwise, before passing it to the PPO update. This phase is the focus of the empirical evaluation in this proposal: it requires no additional training of a safety model, and produces a fast feedback loop that makes the core Minmax claim falsifiable on BeaverTails-Evaluation and HarmBench. Figure 4.1 shows the Phase 1 architecture.

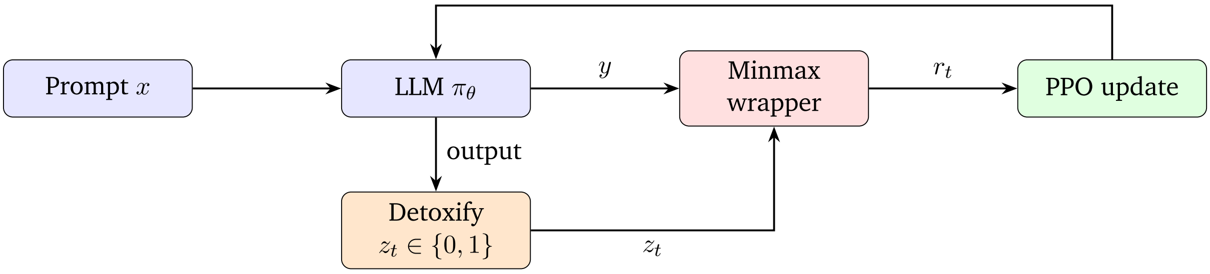

Figure 4.1: Phase 1 architecture. Detoxify classifies the model output, providing the binary unsafe flag z_t to the Minmax wrapper. When $z_t = 1$, the wrapper replaces the reward with $R_{\text{unsafe}} = V_{\text{min}} - V_{\text{max}}$ before passing it to the PPO update.

Phase 2: Activation probe as the unsafe-state detector. Phase 2 keeps the rest of the pipeline fixed and replaces Detoxify with a lightweight activation probe trained on internal model states, following Rateike *et al.* [2023]; Han *et al.* [2025]. The motivation is that surface-level classifiers cannot detect categories such as self-harm, where the output is fluent and non-toxic but the underlying intent is unsafe. Phase 2 is a planned extension scoped as a single milestone; a detailed design is deferred until Phase 1 results are stable. Figure 4.2 shows the Phase 2 architecture.

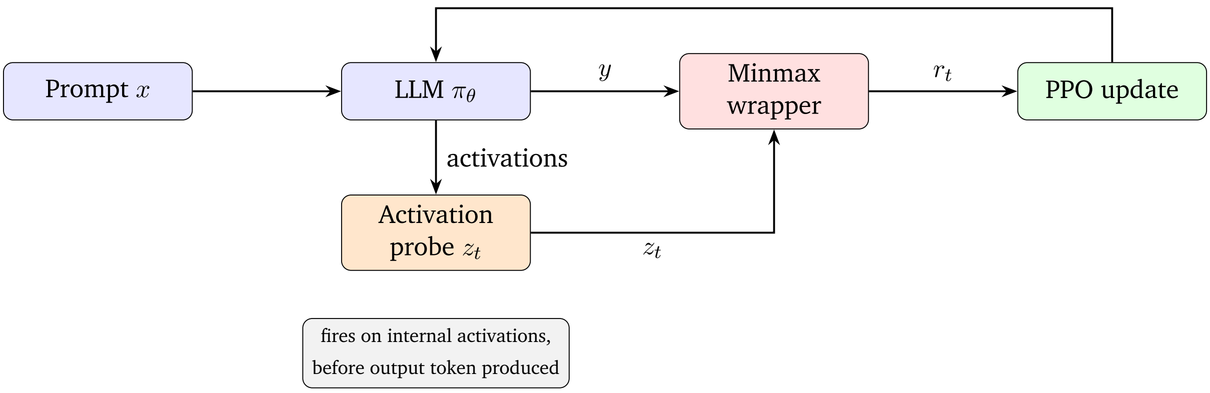

Figure 4.2: Phase 2 architecture. The activation probe replaces Detoxify, detecting unsafe states from internal activations before the output token is produced. This extends Algorithm 1's coverage to harm categories that surface-level classifiers miss.

Evaluation across both phases. Trained policies are evaluated on two complementary benchmarks. The primary, in-distribution benchmark is BeaverTails-Evaluation: 700

held-out prompts across 14 harm categories. Both trained models generate 32 tokens per prompt; outputs are classified as harmful or benign, and harm rate is reported in aggregate and per category. Using the same split as Safe RLHF [Dai *et al.* 2024] and HC-RLHF [Chittepu and others 2025] enables direct comparison with published results.

The secondary, out-of-distribution benchmark is HarmBench [Mazeika *et al.* 2024], which provides over 500 adversarial behaviours designed for red-teaming, spanning standard, contextual, and multi-turn attacks that cross harm categories. HarmBench tests whether the Minmax penalty generalises beyond the naturalistic BeaverTails prompts seen during training, and is the stronger test of H_1: a method that only reduces harm on in-distribution prompts could simply be reproducing the training distribution rather than learning a robust safety policy. Following Mazeika *et al.* [2024], attack success rate (ASR) is the primary metric, reported overall and broken down by attack family. Full HarmBench evaluation is run after Phase 1 results are stable, and again after Phase 2.

At each training step the following are logged: mean reward, $V_{\min}$, $V_{\max}$, R_{unsafe}, policy KL divergence, and PPO ratio statistics, enabling inspection of the self-calibration dynamics predicted by Nangue Tasse and others [2026]. Ablations on safety mechanism, base model, fine-tuning strategy, and detector threshold isolate the contribution of the Minmax mechanism from confounds (see Section 4.1.3).

4.1.3 Ablation Studies

The ablation design is structured to isolate the Minmax mechanism as the causal factor for any observed safety improvement, varying one dimension at a time while holding all others fixed. This is essential to rule out confounds from the choice of base model, fine-tuning strategy, or detector sensitivity.

Ablation 1: Safety mechanism. The core comparison, holding everything fixed and varying only the safety mechanism: PPO with KL penalty (baseline), PPO with Minmax (proposed), and PPO with no safety mechanism (lower bound).

Ablation 2: Base model. Holds the Minmax mechanism fixed and varies the base model between GPT-2 small (hardware-feasible) and a Qwen model at 1 to 3B parameters (cluster). This directly addresses the concern that the observed improvement might be specific to GPT-2 rather than attributable to the Minmax mechanism.

Ablation 3: Fine-tuning method. Holds the Minmax mechanism and base model fixed and varies the fine-tuning strategy between full fine-tuning and LoRA. This addresses the concern that the result might be confounded by the training approach rather than the safety mechanism.

Ablation 4: Detector threshold. Holds all other parameters fixed and varies the unsafe threshold $\tau \in \{0.1, 0.3, 0.5, 0.7\}$ to characterise how detector sensitivity affects harm rate and policy stability.

Extended training runs of 1000 to 2000 steps will assess stability over longer training.

4.1.4 Limitations and Constraints

This research is bounded by several constraints that shape the scope of claims that can be made from its results.

Detector quality bounds coverage. By construction, Algorithm 1 only acts when the unsafe-state detector fires. In Phase 1, this couples the achievable harm reduction to Detoxify's recall: harm categories whose surface form does not register as toxic (most notably self-harm) cannot be improved by the Minmax penalty regardless of how well-tuned the RL loop is. This is the explicit motivation for Phase 2, but it remains a constraint on Phase 1 results.

Benchmark coverage. BeaverTails-Evaluation and HarmBench together cover naturalistic and adversarial harm but do not exhaust the space of safety-relevant behaviours; truthfulness and long-horizon manipulation are not directly measured. HarmBench evaluation depends on access to its judge classifier and is therefore scheduled as a later milestone.

Phase 2 scope. The activation-probe extension is scoped as a single milestone with hallucination as the initial target subset, following Rateike *et al.* [2023]. A full study of probe architectures and layer selection is identified as future work.

Statistical scope. The full experimental programme will use multiple seeds per condition so that reported harm-rate differences can be tested for significance rather than reported as point estimates.

Chapter 5

Research Plan

This chapter outlines the planned experimental phases, target results, success criteria, and proposed future extension.

5.1 Timeline

The planned phases are summarised in Table 5.1, with indicative month ranges relative to the project start in May 2026 and the standard two-year MSc duration. Cluster-dependent phases (Qwen model scale, HarmBench evaluation) are scheduled after cluster access is confirmed.

Table 5.1: Indicative timeline. Month ranges are relative to project start (May 2026).

Phase	Activities	Months	Status
1	Proposal submission and ethics clearance	1–2	In progress
2	Full Phase 1 GPT-2 runs (extended steps, ablations)	2–4	Planned
3	LoRA fine-tuning experiments; threshold ablation	4–5	Planned
4	Qwen model scale experiments (cluster)	5–7	Planned
5	Phase 2 activation probe preliminary experiment	7–9	Planned
6	HarmBench out-of-distribution evaluation	9–10	Planned
7	Thesis writing and submission	10–24	Planned

5.2 Target Results

5.3 Success Criteria

Minimum bar. Minmax training reduces harm rate relative to the KL-penalty baseline on BeaverTails-Evaluation. Extended training shows stable or improving safety performance. Ablation studies document sensitivity to threshold, KL anchor, fine-tuning

Table 5.2: Target results table for thesis. [TBD] entries are planned experiments.

Condition	Harm Rate ↓	Mean Reward ↑	Notes
GPT-2 zero-shot	[TBD]	[TBD]	Untrained baseline
PPO + KL (baseline)	[TBD]	[TBD]	KL-penalty baseline
PPO + Minmax	[TBD]	[TBD]	Proposed method
PPO + Minmax (extended)	[TBD]	[TBD]	Extended training
PPO + Minmax (LoRA)	[TBD]	[TBD]	Fine-tuning ablation
Qwen + Minmax	[TBD]	[TBD]	Model scale ablation
Safe RLHF 7B (from paper)	1–3%	N/A	SOTA [Dai *et al.* 2024]
HC-RLHF (from paper)	[reported]	N/A	Newest SOTA [Chittepu and others 2025]

method, and base model. Per-category analysis attributes coverage gaps (e.g. self-harm under Detoxify) to detector quality.

Target. All minimum criteria, plus: final BeaverTails-Evaluation results competitive with published baselines at comparable model scale; Phase 2 activation probe experiment showing that the self-harm signal is detectable in GPT-2 activations (or providing a clear null result); and HarmBench [Mazeika *et al.* 2024] evaluation for out-of-distribution robustness.

5.4 Proposed Future Extension

Phase 2 is motivated by the expected coverage limits of surface-level detectors: replacing Detoxify with an activation probe trained to detect unsafe internal states before the output token is produced. Based on Rateike *et al.* [2023]'s finding that hallucination detection is the most tractable initial subset, the probe will first target factually incorrect and internally inconsistent outputs. The PKU-SafeRLHF dataset [Ji *et al.* 2024], with its paired safe and unsafe responses across 19 harm categories, provides explicit positive and negative training examples for the probe. Once validated on hallucination detection, the probe will be extended to self-harm and other categories that Detoxify misses, directly answering the open question posed by Cintas *et al.* [2025].

Chapter 6

Conclusion

This proposal presents a research programme investigating whether the ROSARL Minmax penalty can serve as the primary safety mechanism in PPO-based LLM fine-tuning, replacing both KL divergence penalties and learned cost models. The central motivation is that the RL loop already contains the information needed for safe behaviour, in the form of the actor-critic algorithm's own value function, and that adding external mechanisms may not be necessary.

The research direction follows from the structure of Algorithm 1: the penalty mechanism acts only when the unsafe-state detector fires, so detector quality is the primary variable modulating coverage. The two-phase architecture, moving from surface-level toxicity detection to activation-based safety signals, is designed to address this directly. The planned ablation programme, varying base model, fine-tuning strategy, and detector threshold, will establish whether observed improvements are attributable to the Minmax mechanism rather than to any specific experimental configuration.

A successful thesis outcome, completing both phases, confirming mechanism-specificity through the ablations, and demonstrating that Phase 2 extends coverage to categories such as self-harm, would contribute to the safe RLHF literature a concrete demonstration that value-function-derived signals can replace both KL penalties and learned cost models, and that the binding constraint on safety coverage is detector quality rather than algorithm design. This places the Minmax penalty within the emerging RL with verifiable rewards paradigm [Lambert and others 2024; Shao *et al.* 2024; Huang and others 2025]: the signal is computed deterministically from the agent's own training trajectory rather than approximated by a learned model. For practitioners, the approach is attractive because it requires no additional annotation, no separate trained model, and no new safety-specific hyperparameters, making it accessible to teams working within tight compute budgets and providing a path toward safer PPO-based fine-tuning that does not depend on access to large annotation pipelines.

References

[Achiam *et al.* 2017] Joshua Achiam, David Held, Aviv Tamar, and Pieter Abbeel. Constrained policy optimization. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017.

[Bai *et al.* 2022] Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.

[Chaudhary and Barez 2025] Maheep Chaudhary and Fazl Barez. SafetyNet: Detecting harmful outputs in LLMs by modeling and monitoring deceptive behaviors. *arXiv preprint arXiv:2505.14300*, 2025.

[Chittepu and others 2025] Sai Chittepu et al. HC-RLHF: Reinforcement learning from human feedback with high-confidence safety constraints. *arXiv preprint arXiv:2506.08266*, 2025.

[Christiano *et al.* 2017] Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[Cintas *et al.* 2025] Celia Cintas, Miriam Rateike, Erik Miehling, Elizabeth Daly, and Skyler Speakman. Localizing persona representations in LLMs. In *AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 2025.

[Dai *et al.* 2024] Josef Dai, Xuehai Pan, Ruiyang Sun, Jiaming Ji, Xinbo Xu, Mickel Liu, Yizhou Wang, and Yaodong Yang. Safe RLHF: Safe reinforcement learning from human feedback. In *International Conference on Learning Representations (ICLR)*, 2024.

[García and Fernández 2015] Javier García and Fernando Fernández. A comprehensive survey on safe reinforcement learning. *Journal of Machine Learning Research*, 16:1437–1480, 2015.

[Han *et al.* 2025] Peixuan Han, Cheng Qian, Xiusi Chen, Yuji Zhang, Denghui Zhang, and Heng Ji. SafeSwitch: Steering unsafe LLM behavior via internal activation signals. In *Findings of EMNLP*, 2025.

[Hanu and Unitary team 2020] Laura Hanu and Unitary team. *Detoxify*. `https://github.com/unitaryai/detoxify`, 2020.

[Hu *et al.* 2022] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. *International Conference on Learning Representations (ICLR)*, 2022. arXiv:2106.09685.

[Huang and others 2025] Farrel Huang et al. Breaking the safety–capability trade-off: Reinforcement learning with verifiable rewards maintains safety guardrails in LLMs. *arXiv preprint arXiv:2511.21050*, 2025. Author list to be verified against arXiv:2511.21050 before submission.

[Ji *et al.* 2023] Jiaming Ji, Shuai Liu, Josef Dai, et al. BeaverTails: Towards improved safety alignment of LLM via a human-preference dataset. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[Ji *et al.* 2024] Jiaming Ji, Donghai Hong, Borong Zhang, Boyuan Chen, Josef Dai, Boren Zheng, Tianyi Qiu, Boxun Li, and Yaodong Yang. PKU-SafeRLHF: A safety alignment preference dataset for Llama family models. In *arXiv preprint arXiv:2406.15513*, 2024.

[Lambert and others 2024] Nathan Lambert et al. Reinforcement learning with verifiable rewards. *arXiv preprint*, 2024. arXiv:2408.02232.

[Lambert *et al.* 2023] Nathan Lambert, Thomas K. Gilbert, and Tom Zick. Entangled preferences: The history and risks of reinforcement learning and human feedback. *arXiv preprint arXiv:2310.13595*, 2023.

[Lockwood *et al.* 2020] Patricia L. Lockwood, Miriam C. Klein-Flügge, Ayat Abdurahman, and Molly J. Crockett. Model-free decision making is prioritized when learning to avoid harming others. *Proceedings of the National Academy of Sciences*, 117(44):27719–27730, 2020.

[Mazeika *et al.* 2024] Mantas Mazeika, Long Phan, Xuwang Yin, et al. HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. In *International Conference on Machine Learning (ICML)*, 2024.

[Mnih *et al.* 2015] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, et al. Human-level control through deep reinforcement learning. *Nature*, 518:529–533, 2015.

[Nangue Tasse and others 2026] Geraud Nangue Tasse et al. An unreasonably simple approach to safe reinforcement learning. RLC 2026 (under review), 2026.

[Nguyen *et al.* 2025] Bao Nguyen, Binh Nguyen, Duy Nguyen, and Viet Nguyen. Distributional surgery for language model activations. *Findings of EMNLP*, 2025.

[Orban and others 2024] Gergo Orban et al. NeuroAI for AI safety. *arXiv preprint arXiv:2411.18526*, 2024.

[Ouyang *et al.* 2022] Long Ouyang, Jeffrey Wu, Xu Jiang, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[Puterman 2014] Martin L. Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. John Wiley & Sons, 2014.

[Rateike *et al.* 2023] Miriam Rateike, Celia Cintas, John Wamburu, Michael Akumu, and Skyler Speakman. Weakly supervised detection of hallucinations in LLM activations. In *NeurIPS Workshop on Socially Responsible Language Modelling Research (SoLaR)*, 2023.

[Schulman *et al.* 2017] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[Shao *et al.* 2024] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

[Silver *et al.* 2016] David Silver, Aja Huang, Chris J. Maddison, et al. Mastering the game of Go with deep neural networks and tree search. *Nature*, 529:484–489, 2016.

[Su and others 2025] Yi Su et al. Crossing the reward bridge: Expanding RL with verifiable rewards across diverse domains. *arXiv preprint*, 2025. arXiv:2503.23829.

[Sutton and Barto 2018] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2nd edition, 2018.

[von Werra *et al.* 2020] Leandro von Werra, Younes Belkada, Lewis Tunstall, Edward Beeching, Tristan Thrush, Nathan Lambert, and Shengding Hu. *TRL: Transformer Reinforcement Learning*. https://github.com/huggingface/trl, 2020.

[Yang *et al.* 2022] Qisong Yang, Thiago D. Simão, Simon H. Tindemans, and Matthijs T. J. Spaan. Safety-constrained reinforcement learning with a distributional safety critic. *Machine Learning*, 2022.

[Youssef and others 2025] Ahmed Youssef et al. Reinforcement learning with verifiable rewards: GRPO's effective loss, dynamics, and success amplifications. In *arXiv preprint*, 2025. arXiv:2503.01491.

[Ziegler *et al.* 2019] Daniel M. Ziegler, Nisan Stiennon, Jeffrey Wu, Tom B. Brown, Alec Radford, Dario Amodei, Paul Christiano, and Geoffrey Irving. Fine-tuning language models from human preferences. *arXiv preprint arXiv:1909.08593*, 2019.